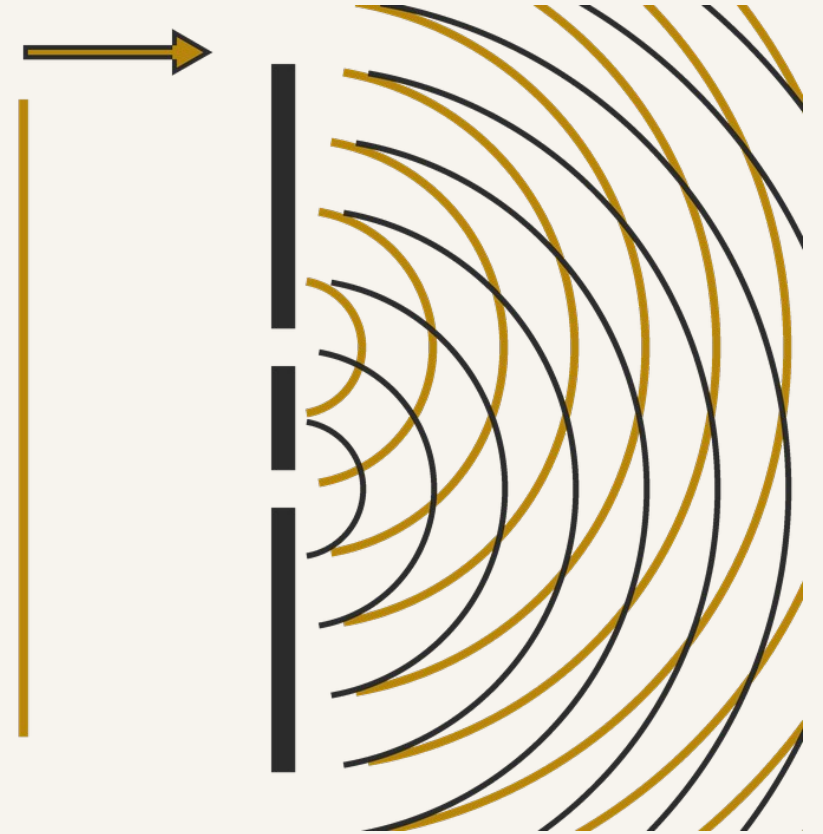


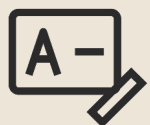
Division of Wavefront

Interference part 1: two slits, and a pattern that proved light is a wave.

Advanced Higher Physics



What the pictures mean



Show me boards
Hold up your board.



Write it
Write it in your jotter.



Turn and talk
Talk to your partner.



Hands down
No hands up. I pick.



Hands up
Hand up to answer.



Silent task
Work alone, in silence.



Jotter notes
Copy it into your jotter.



Cut and stick
Cut out and stick in.

LEARNING INTENTION

To explain how coherent waves interfere, and to use Young's double slits to measure the wavelength of light.

By the end of this deck...

- 1 I can state what is meant by coherent waves.
- 2 I can state the conditions for constructive and destructive interference in terms of phase and path difference.
- 3 I can explain interference by division of wavefront.
- 4 I can use $\Delta x = \lambda D/d$ to solve problems, including uncertainties.

Daily Review



- 1 State what is meant by the wavelength of a wave.
- 2 What is meant by monochromatic light?
- 3 Two waves meet in phase. What happens to the amplitude?
- 4 State the relationship linking speed, frequency and wavelength.
- 5 At Higher, what path difference gives a maximum?



Two stones drop into a calm lake. Why are some places still calm?

PART 1

When do two waves add up?

Only when they keep in step.

By the end of this part...

- 1 I can state what is meant by coherent waves.
- 2 I can state the conditions for constructive and destructive interference.
- 3 I can link path difference to phase difference.

Coherent waves



LEARN THIS WORD FOR WORD

Two waves are coherent if they have a
constant phase relationship.

Not coherent – a lamp



A lamp

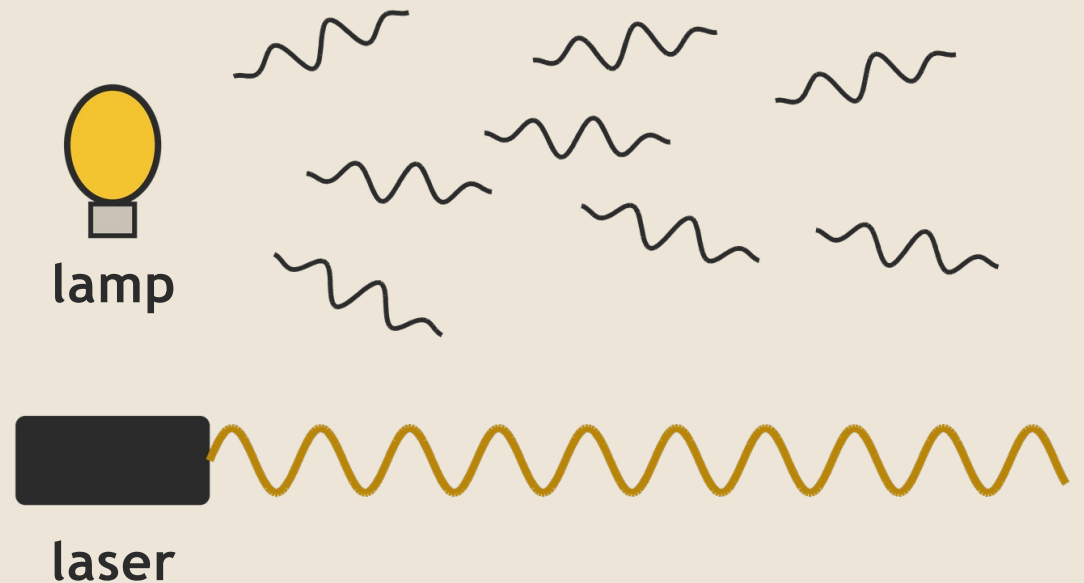
Atoms emit short random bursts.

Random phase

No constant phase relationship.

A laser

One long train: coherent.



Not coherent – two frequencies



Two frequencies

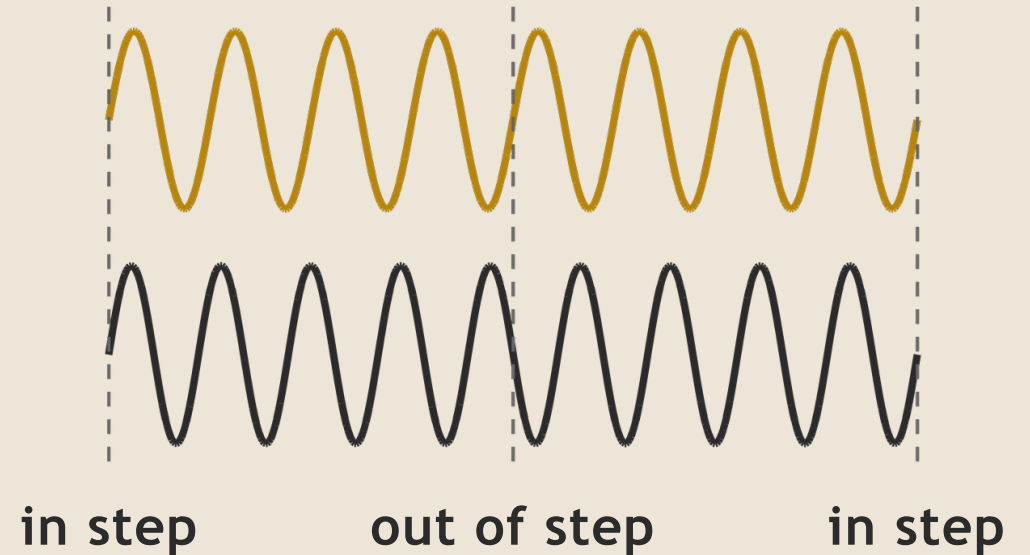
They drift in and out of step.

Phase gap

Keeps changing.

Coherent needs

The same frequency.

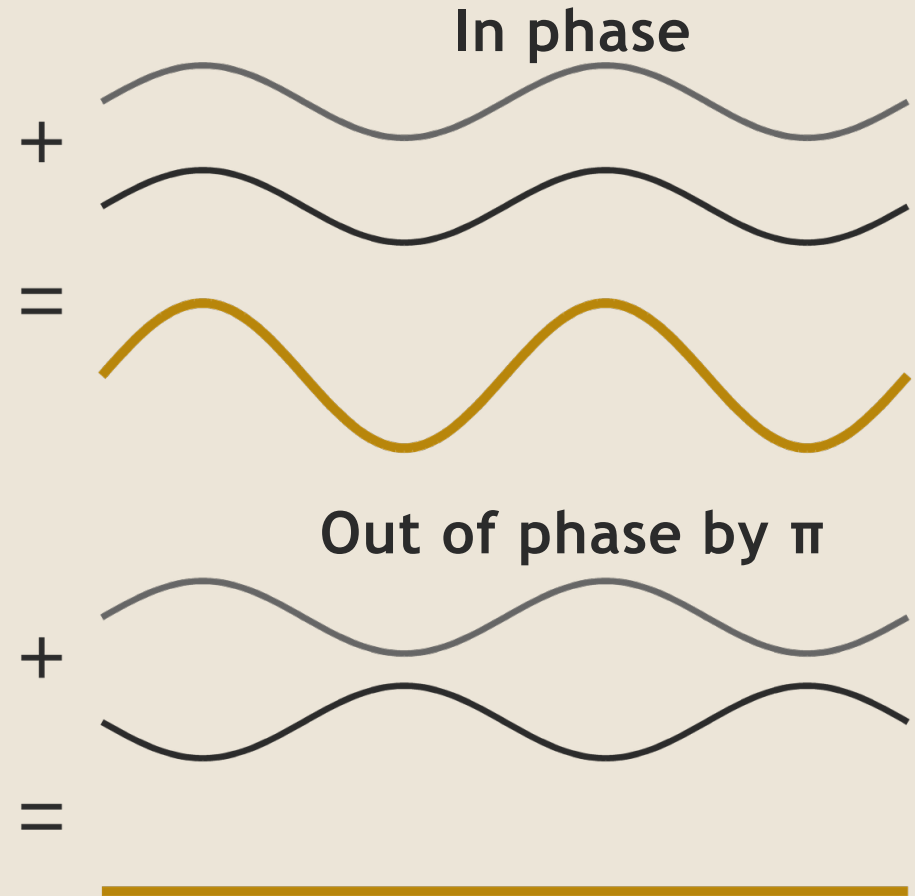


Constructive and destructive



In phase: they add. A maximum.

Out of phase by π : they cancel. A minimum.



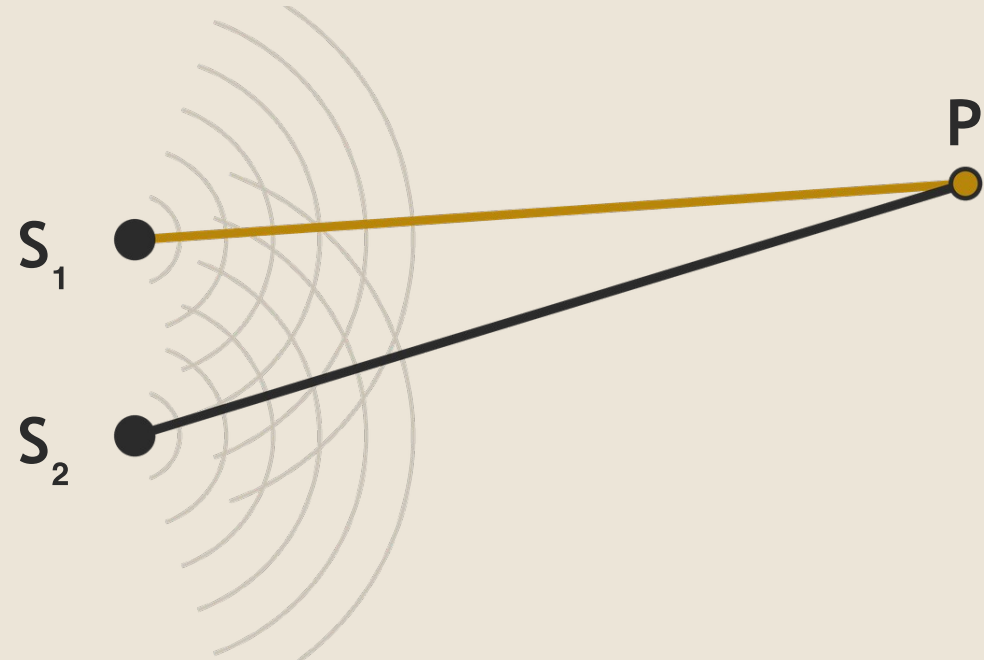
Path difference



Maximum: path difference
 $m\lambda$

Minimum: path difference
 $(m + \frac{1}{2})\lambda$

$m = 0, 1, 2 \dots$



$$\text{Path difference} = S_2P - S_1P$$

Phase difference



$$\phi = 2\pi x / \lambda$$

ϕ : phase difference (rad)

x : path difference (m)

λ : wavelength (m)

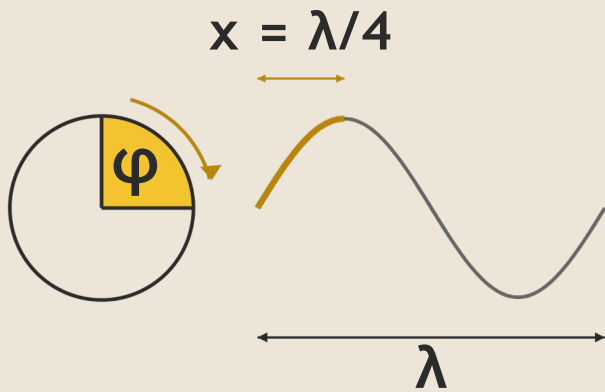
One wavelength is 2π , so
 $\lambda/4$ is $\pi/2$.



Phase difference – other ways to see it



Same fraction



One turn = 2π



$$x = 0: \varphi = 0$$



$$x = \lambda/4: \varphi = \pi/2$$



$$x = \lambda/2: \varphi = \pi$$



$$x = \lambda: \varphi = 2\pi$$

Four cases

Look back — by the end of this part...

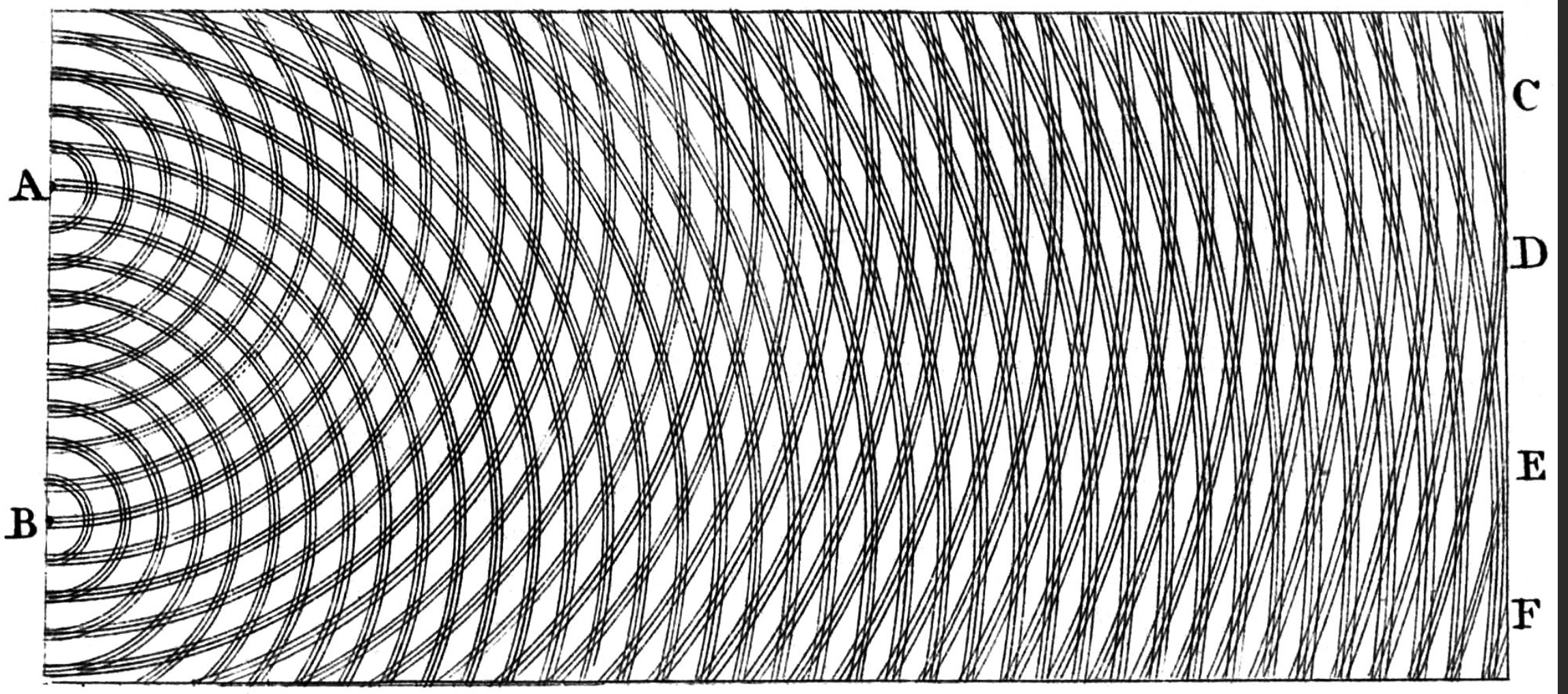


- 1 I can state what is meant by coherent waves.
- 2 I can state the conditions for constructive and destructive interference.
- 3 I can link path difference to phase difference.

Daily Review



- 1 State what is meant by coherent waves.
- 2 State the condition for destructive interference.
- 3 Why is light from a lamp not coherent?
- 4 Waves meet out of phase by π . What is seen?
- 5 Write the path difference for the second maximum.



Thomas Young drew this in 1803. What did it prove about light?

About Thomas Young

Who

Doctor and physicist.

Born

1773, Milverton, Somerset.

Died

1829, London.

Languages

Learned many as a boy.



Thomas Young – discoveries and theories

Light

His fringes: light is a wave.

Colour vision

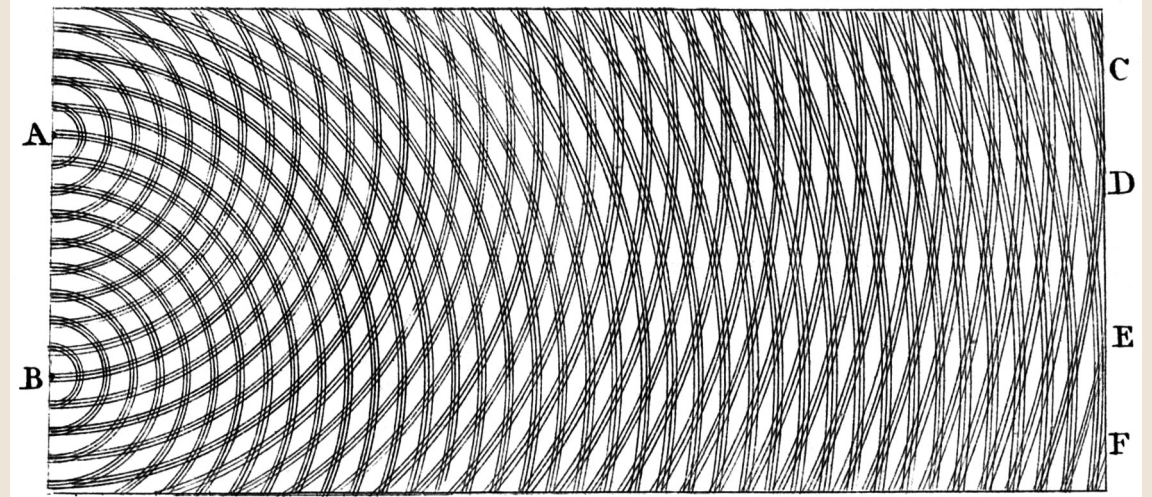
Eye has 3 colour receptors.

Rosetta Stone

Helped decode hieroglyphs.

Young's modulus

How stiff a material is.



PART 2

How do you split one wave into two?

Let one wavefront through two slits.

By the end of this part...

- 1 I can explain interference by division of wavefront.
- 2 I can describe Young's double slit experiment.
- 3 I can use $\Delta x = \lambda D/d$ to solve problems.

Division of wavefront



Sodium lamp

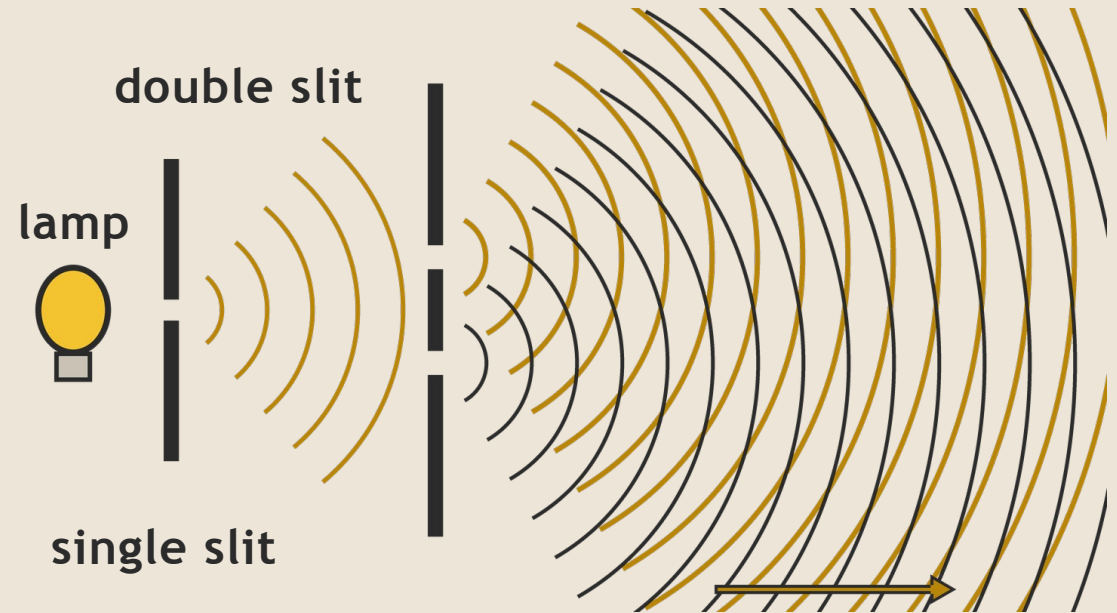
Not coherent on its own.

Single slit first

Makes one single wavefront.

Double slit

Splits it into two coherent sources.



Young's double slits

No single slit

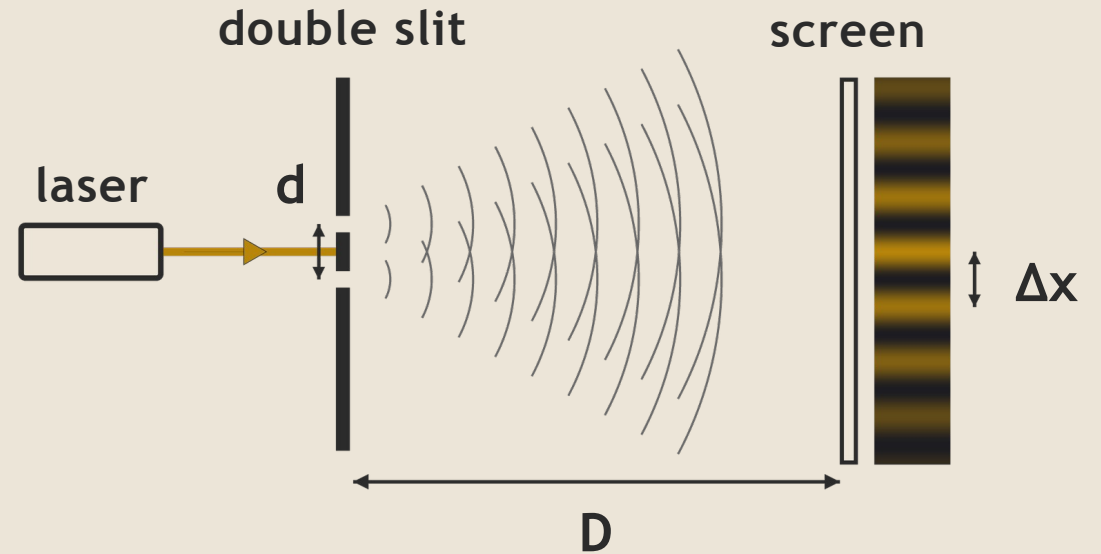
A laser is already coherent.

One wavefront

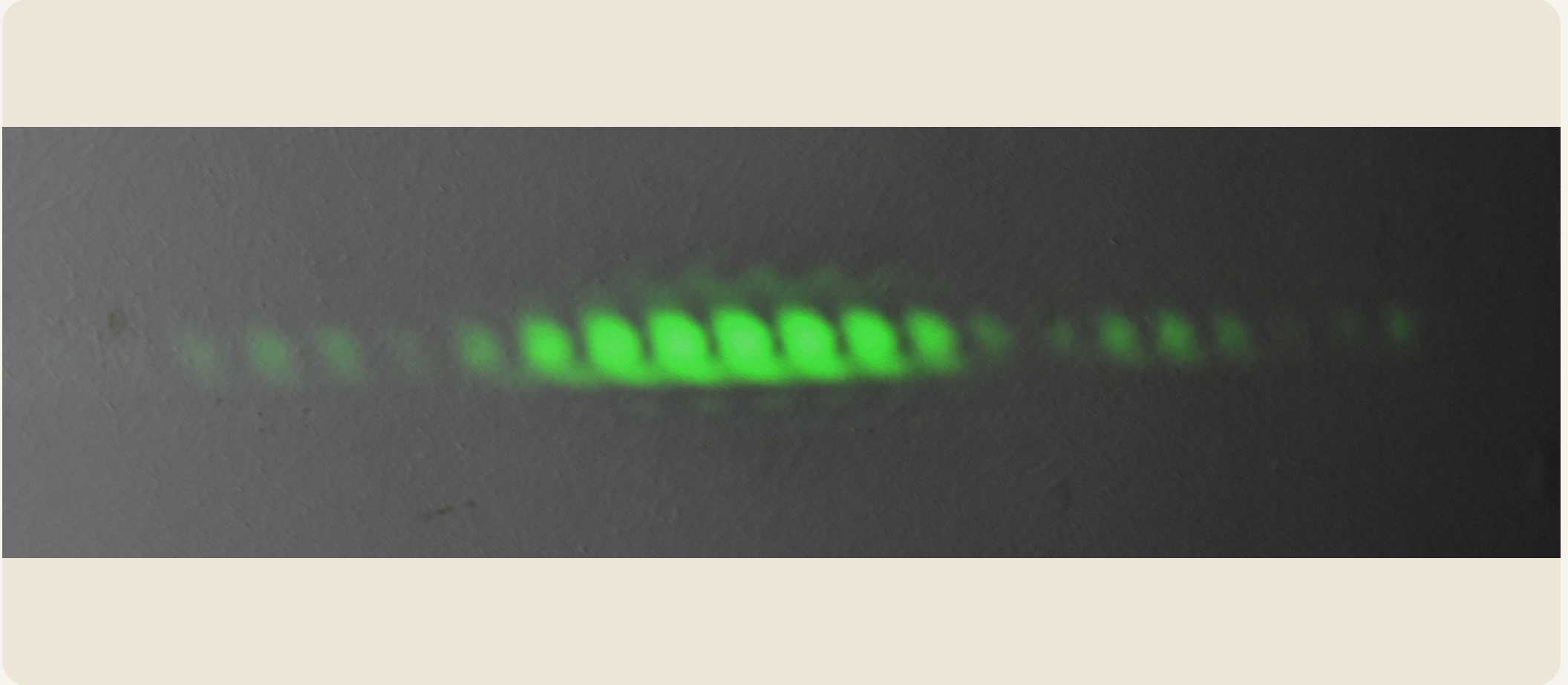
The beam lights both slits.

Δx

Bright fringe to the next one.



A real Young's slits pattern



Green laser, slits 0.1 mm wide and 0.4 mm apart.

Fringe separation



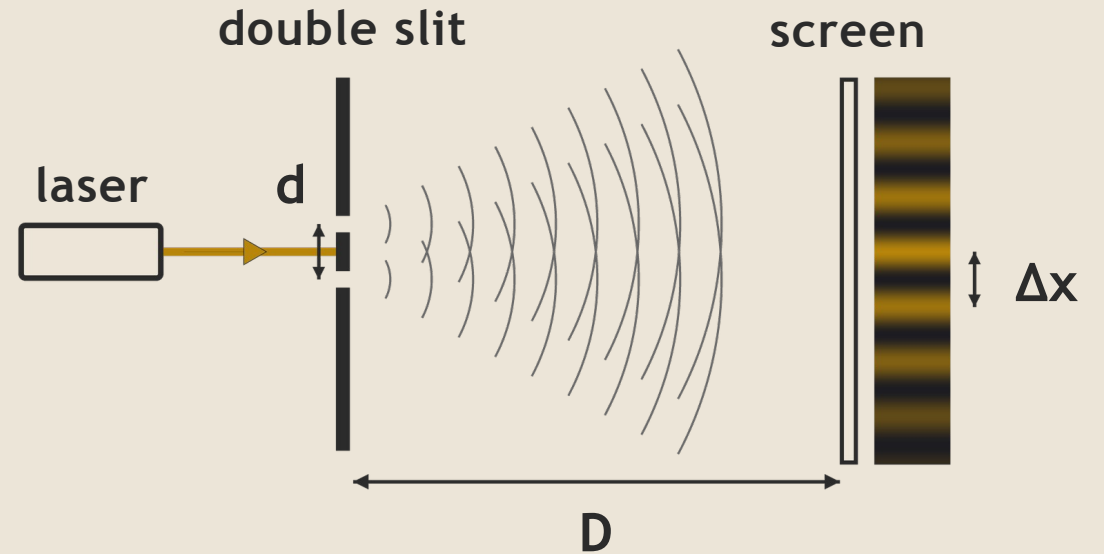
$$\Delta x = \lambda D / d$$

Δx : fringe separation (m)

λ : wavelength (m)

D : slits to screen (m)

d : slit separation (m)



What changes the fringe separation?



Δx rises with λ and with D .

Δx falls as d rises.

Answer in two parts: the effect, then why.

Why: what changed, and what stayed the same.

red



blue



Same slits, same D . Red light has the longer wavelength.

Example answer – change λ

A blue laser replaces the green laser.

State the effect on the fringe separation.

You must justify your answer. (2 marks)

blue: shorter λ

d unchanged

D unchanged

Fringe separation decreases. (1)

λ decreases; d, D unchanged. (1)

No reason given: 0 marks.



Example answer – change D

The screen is moved further from the slits.

State the effect on the fringe separation.

You must justify your answer. (2 marks)

D increases

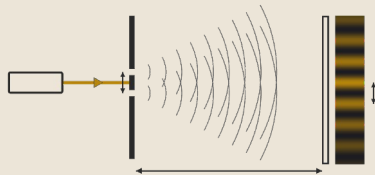
λ unchanged

d unchanged

Fringe separation increases. (1)

D increases; λ , d unchanged. (1)

No reason given: 0 marks.



Example answer – change d

The slits are replaced by slits closer together.

State the effect on the fringe separation.

You must justify your answer. (2 marks)

d decreases

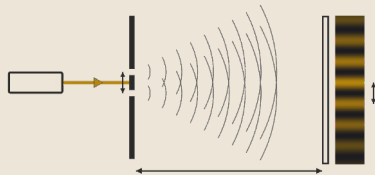
λ unchanged

D unchanged

Fringe separation increases. (1)

d decreases; λ , D unchanged. (1)

No reason given: 0 marks.



Worked example 1 – fringe separation

A red laser of wavelength 650 nm lights a double slit of slit separation 0.25 mm. The screen is 2.80 m away. Calculate Δx .

$$\lambda = 650 \times 10^{-9} \text{ m}$$

$$d = 0.25 \times 10^{-3} \text{ m}$$

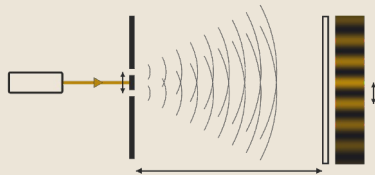
$$D = 2.80 \text{ m}$$

$$\Delta x = ?$$

$$\Delta x = \lambda D / d$$

$$\Delta x = 650 \times 10^{-9} \times 2.80 / 0.25 \times 10^{-3}$$

$$\underline{\Delta x = 7.28 \times 10^{-3} \text{ m}}$$



Worked example 2 – across many fringes

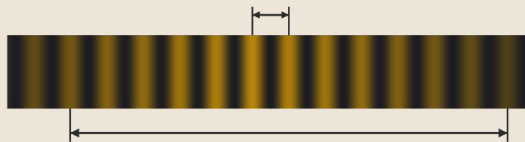
A pupil measures 45.6 mm across 12 fringe separations. The slits are 0.50 mm apart and 3.40 m from the screen. Calculate λ .

$$12\Delta x = 45.6 \times 10^{-3} \text{ m}$$

$$d = 0.50 \times 10^{-3} \text{ m}$$

$$D = 3.40 \text{ m}$$

$$\lambda = ?$$



$$\Delta x = 45.6 \times 10^{-3} \div 12 = 3.80 \times 10^{-3} \text{ m}$$

$$\Delta x = \lambda D / d$$

$$3.80 \times 10^{-3} = \lambda \times 3.40 / 0.50 \times 10^{-3}$$

$$\underline{\lambda = 5.59 \times 10^{-7} \text{ m}}$$

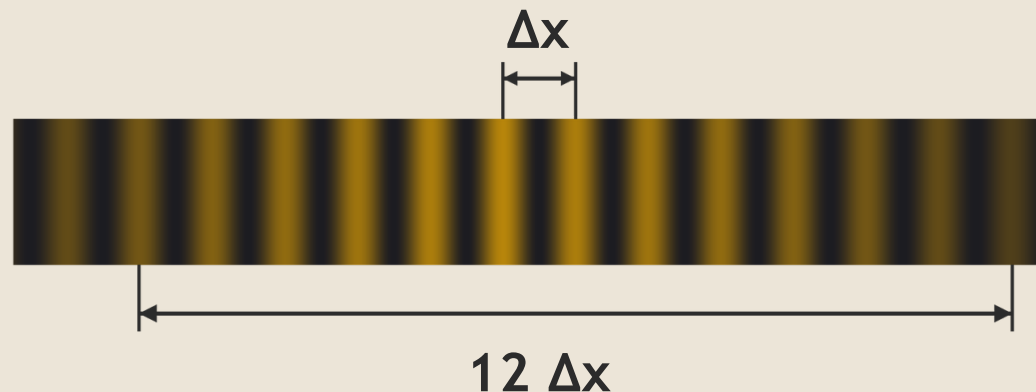
Measuring Δx well



Measure across many fringes, then divide.

Smaller percentage uncertainty in Δx .

Moving the screen back does the same.



$$\frac{\Delta(\Delta x)}{\Delta x} = \sqrt{\left(\frac{\Delta D}{D}\right)^2 + \left(\frac{\Delta d}{d}\right)^2}$$

Worked example 3 – uncertainty in Δx

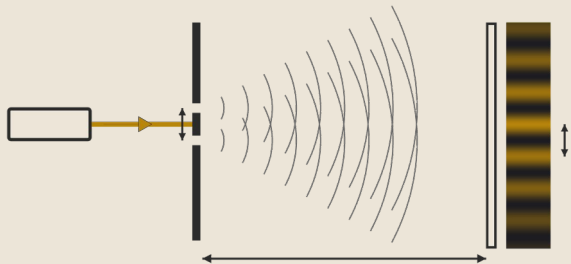
In example 1, $D = (2.80 \pm 0.02) \text{ m}$ and $d = (0.25 \pm 0.01) \text{ mm}$.

Calculate the absolute uncertainty in $\Delta x = 7.28 \times 10^{-3} \text{ m}$.

$$\Delta D/D = 0.02 \div 2.80$$

$$\Delta d/d = 0.01 \div 0.25$$

$$\Delta(\Delta x) = ?$$



$$\Delta(\Delta x)/\Delta x = \sqrt{((\Delta D/D))^2 + ((\Delta d/d))^2}$$

$$\Delta(\Delta x)/7.28 \times 10^{-3} = \sqrt{(0.0071)^2 + (0.040)^2}$$

$$\Delta(\Delta x) = 3 \times 10^{-4} \text{ m}$$

$$\underline{\Delta x = (7.3 \pm 0.3) \times 10^{-3} \text{ m}}$$

Where you meet it — radio telescopes



Dishes far apart catch the same wavefront. Combined, they see as much detail as one dish 36 km wide.

For interest – how the dishes see detail

One wavefront

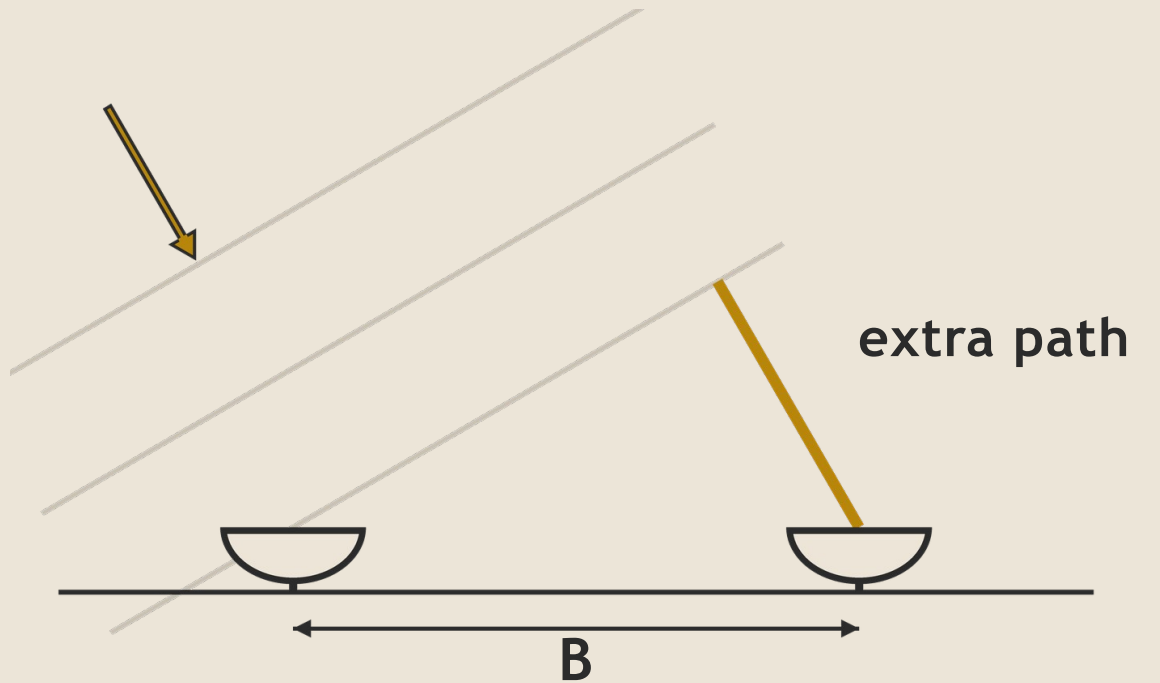
Reaches one dish first.

Path difference

Changes as the source moves.

Far apart

A tiny move changes it a lot.

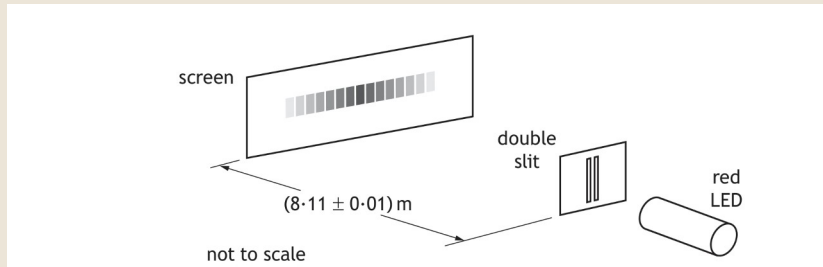


Look back — by the end of this part...

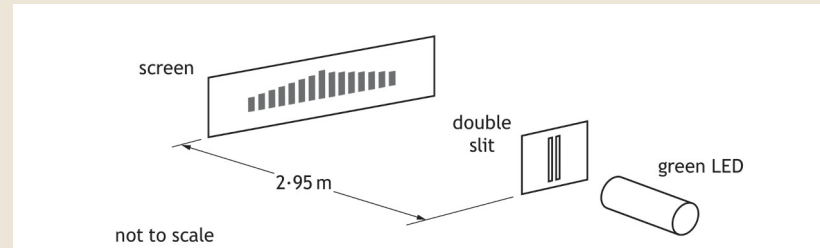


- 1 I can explain interference by division of wavefront.
- 2 I can describe Young's double slit experiment.
- 3 I can use $\Delta x = \lambda D/d$ to solve problems.

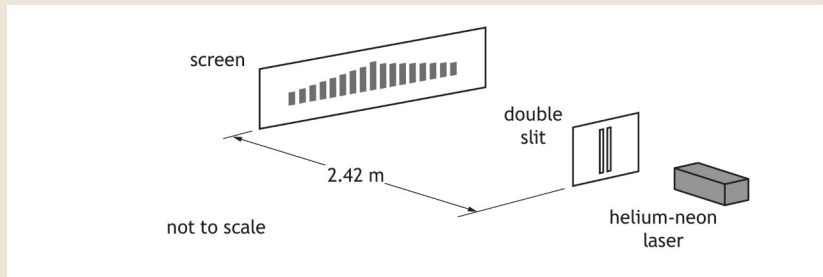
Division of wavefront in the exam



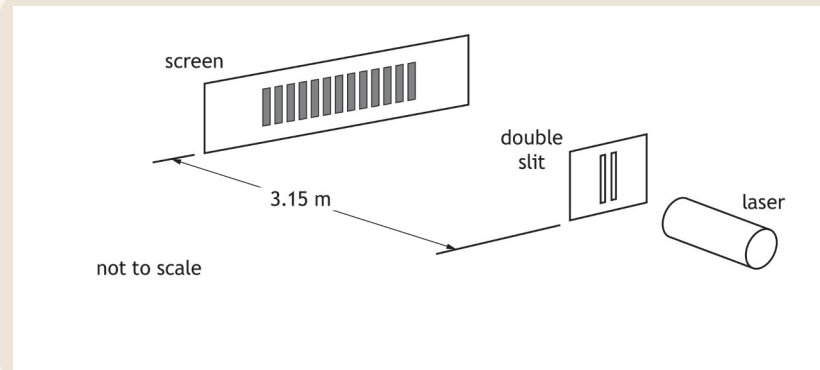
2017 Q10(c)



2019 Q13(a), (b)



2022 Q12



2024 Q11(a), (b)



Past paper
finder

2025 Q11 is kept
for your test.

Reading and homework



Scholar Topic 8 test

Scholar – Topic 6 Interference

Scholar – Topic 8 Division of wavefront,
end-of-topic test

BrightRed Study Guide pp. 80-81 and 84-
85

Past papers – 2017, 2019, 2022, 2024



Scholar

Errors that cost marks

- "Increases" or "decreases" alone scores 0 – justify it.
- Divide by the number of fringe separations, not fringes.
- Convert mm and nm to metres before substituting.
- d is the slit separation, not the slit width.
- Say percentage uncertainty, never scale reading uncertainty.

Check yourself

- 1 State what is meant by coherent waves, word for word.
- 2 State the path difference for a bright and a dark fringe.
- 3 Explain interference by division of wavefront.
- 4 Find λ from a measurement across N fringe separations.
- 5 State and justify what happens to Δx when λ , D or d changes.

Experiments you could try

- Class 2 laser and a double slit: measure λ . Never look into the beam.
- Microwave transmitter and metal plates: find the maxima.
- Two loudspeakers on one signal generator: walk across the room.
- Ripple tank with two dippers on one bar.
- Photograph your fringes with a ruler in shot and measure Δx .

Course specification – 1 of 2

- Knowledge that two waves are coherent if they have a constant phase relationship.
- Knowledge of the conditions for constructive and destructive interference in terms of coherence and phase.

Course specification – 2 of 2

- Explanation of interference by division of wavefront.
- Knowledge of Young's slits interference.
- Use of an appropriate relationship to solve problems involving interference of waves by division of wavefront.
- $\Delta x = \lambda D / d$

Credits (1 of 2)

Part 1 hook: Photo: Markus Pössel, CC BY 4.0, Wikimedia Commons

Coherent waves: © Qualifications Scotland (SQA)

Part 2 hook: Thomas Young (1803), public domain, Wikimedia Commons

About Thomas Young: Engraving by Henry Adlard after Thomas Lawrence, public domain, Smithsonian Libraries, Wikimedia Commons

Thomas Young – discoveries and theories: Thomas Young (1803), public domain, Wikimedia Commons

A real Young's slits pattern: Photo: Graham Beards, CC BY-SA 4.0, Wikimedia Commons

Where you meet it – radio telescopes: Photo: Dameon Hudson, CC BY 3.0, Wikimedia Commons

Division of wavefront in the exam: © Qualifications Scotland (SQA). AH Physics 2017, 2019, 2022, 2024.

Credits (2 of 2)

Reading and homework: SCHOLAR AH Physics © Heriot-Watt University. BrightRed AH Physics Study Guide

Course specification – 1 of 2: © Qualifications Scotland (SQA). Course spec v2.0, pp.16-17

Course specification – 2 of 2: © Qualifications Scotland (SQA). Course spec v2.0, pp.16-17